

Unit 5, Lesson 4: The Triangle Inequality Theorem and the Converse of the Pythagorean Theorem

Benchmark

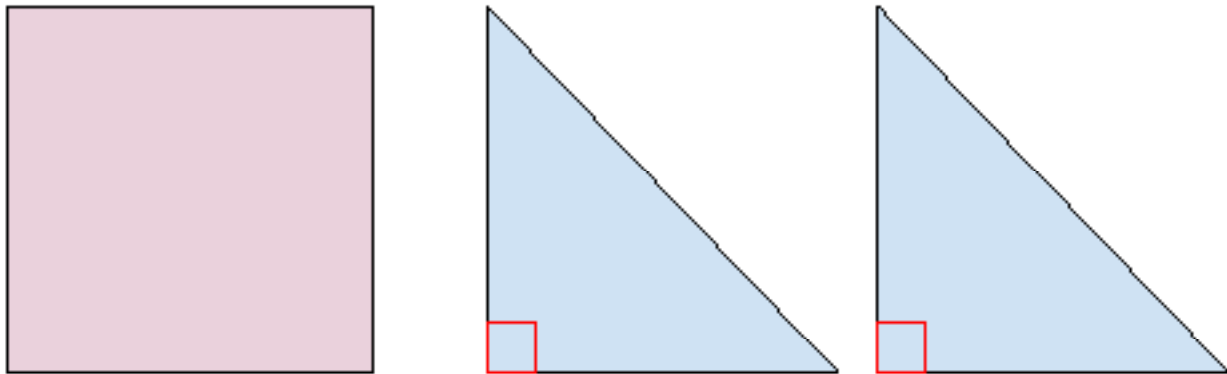
MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Learning Targets

- ☐ I can use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides.
- ☐ I can use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

5.4.1 Warm-Up: Notice and Wonder

1. A square and two right triangles are shown.



- a. What do you notice about the square and the two triangles?
- b. What do you wonder about the square and the two triangles?

5.4.2 Exploration: Triangle Inequality Theorem

- 1. You will receive five different lengths of string, paper, or straw for this activity.
 - a. Attempt to make three different triangles on your desk using different sizes for the sides.
 - b. Measure each side and record the lengths in the table. Indicate if the chosen side lengths formed a triangle or not.

Attempt	Side Length 1	Side Length 2	Side Length 3	Triangle Formed?
A				☐ Yes ☐ No
B				☐ Yes ☐ No
C				☐ Yes ☐ No

- c. For each attempt, compare the sum of two side lengths to the length of the third.
- d. Record the results in the table.

	Attempt A	Attempt B	Attempt C
side 1 + side 2 > side 3	☐ True ☐ False	☐ True ☐ False	☐ True ☐ False
side 1 + side 3 > side 2	☐ True ☐ False	☐ True ☐ False	☐ True ☐ False
side 2 + side 3 > side 1	☐ True ☐ False	☐ True ☐ False	☐ True ☐ False

2. Complete the statements.

a. When the chosen side lengths formed a triangle, the side length

comparisons were

- ☐ all true.
- ☐ all false.
- ☐ not all true.

b. When the chosen side lengths did not form a triangle, the side length

comparisons were

- ☐ all true.
- ☐ all false.
- ☐ not all true.



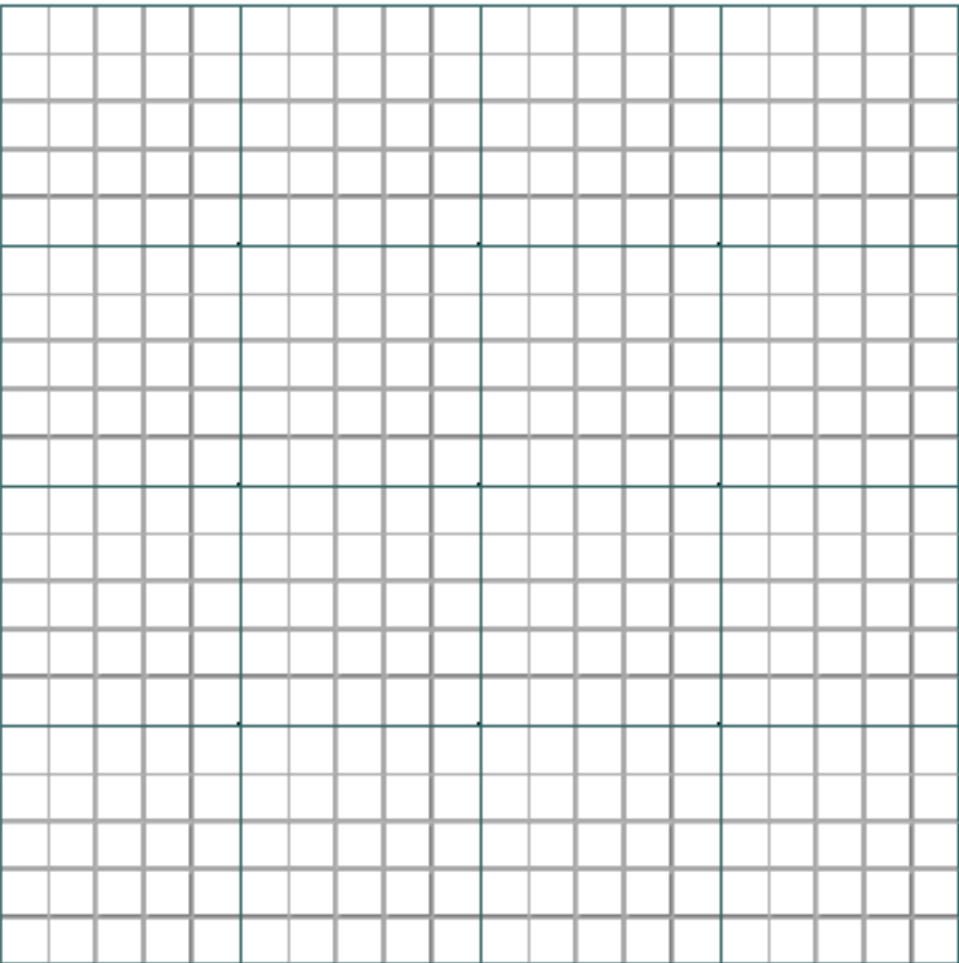
Triangle Inequality Theorem

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

3. Draw a picture to show what happens when the length of the third side of a triangle is longer than the sum of the other two sides.

5.4.3 Guided Instruction: Converse of the Pythagorean Theorem

1. Use the grid to complete the steps.



- In the middle of the grid, draw a right triangle with legs 3 units long and 4 units long. Connect the legs to form the hypotenuse that is 5 units.
- Draw a square with side lengths of 3 units on the outside of the triangle and that shares a side with a leg of the triangle.
- Draw a square with side lengths of 4 units on the outside of the triangle and that shares a side with the other leg of the triangle.
- Draw a square with side lengths of 5 units on the outside of the triangle and that shares a side with the hypotenuse of the triangle.

2. Complete the table with the area of each square.

Side Length (units)	Area of Square (square units)
3	
4	
5	

3. Complete the comparison for the sum of the areas of the squared legs to the square of the hypotenuse.

$3^2 + 4^2 \square 5^2$

5.4.4 Collaboration: The Triangle Inequality Theorem and the Converse of the Pythagorean Theorem

You will receive five different lengths of string, paper, or straw for this activity.

1. Discuss with your partner if a triangle can be formed with the side lengths of 2 centimeters (cm), 4 cm, and 5 cm.

a. Test your prediction by attempting to form a triangle with the side lengths of 2 cm, 4 cm, and 5 cm.
- b. Discuss with your partner if the given lengths form a triangle. If the lengths formed a triangle, classify the type of triangle. If the lengths did not form a triangle, explain why not. Summarize your discussion.
- c. Complete the comparison for the sum of the areas of the squared legs to the square of the hypotenuse.

$2^2 + 4^2 \square 5^2$

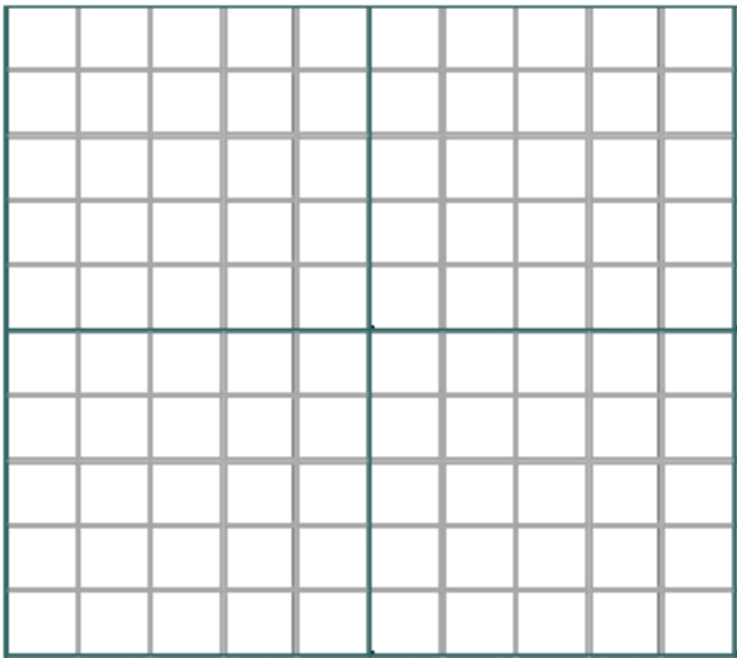
2. Discuss with your partner if a triangle can be formed with the side lengths of 6 cm, 8 cm, and 10 cm.

a. Complete the comparison for the sum of the areas of the squared legs to the square of the hypotenuse.

$6^2 + 8^2 \square 10^2$

b. Based on the comparison, predict the classification of the triangle formed with side lengths of 6 cm, 8 cm, and 10 cm.

c. Sketch a triangle with side lengths of 6 cm, 8 cm, and 10 cm.



d. What is the classification the triangle formed?

e. Discuss with your partner whether your prediction was correct or incorrect. Include how the comparison of the square of the legs to the square of the hypotenuse helped with your prediction. Summarize your discussion.



The **Converse of the Pythagorean Theorem** states that if the lengths a , b , and c of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$ then the triangle is a right triangle.

5.4.5 Your Turn!

1. Determine if each set of side lengths form a triangle. If the side lengths form a triangle, determine if it is a right triangle.

Side Lengths	Triangle?	Right Triangle?
6 in., 12 in., and 13 in.	☐ Yes ☐ No	☐ Yes ☐ No
4 ft, 15 ft, and 17 ft	☐ Yes ☐ No	☐ Yes ☐ No
5 yd, 10 yd, and 15 yd	☐ Yes ☐ No	☐ Yes ☐ No

2. Pablo and Gloria were given the sides lengths 5 cm, 12 cm, and 17 cm then asked to determine if a right triangle was formed.

Pablo says it is a right triangle because $5 + 12 = 17$.	Gloria says it is not a right triangle because $5^2 + 12^2 \neq 17^2$.
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- a. Explain why Gloria is correct.
- b. What was Pablo’s mistake or misconception?
- c. What does Pablo’s statement tell you about the given side lengths?

5.4.6 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest for today’s learning target.

